Response to the reviewers (From Magic State Distillation to Dynamical Systems)

We denote revised part using red color.

Reviewer 1

Reviewer Point 1.1 — Summary: what are the main questions posed by the manuscript and how does it answer them?

This question develops tools to analyse magic state distillation protocols using techniques from dynamical systems theory, in particular, deriving recurrence relations which describe the effect of a single successful magic state iteration on a Bloch sphere vector, and then analysing this using dynamical systems techniques to derive fixed points and flows.

Reply: We thank the reviewer for their time and effort on reviewing and summarizing our manuscript.

Reviewer Point 1.2 — What is your assessment of the paper? If you recommend acceptance, make a case that this work does indeed make a significant contribution to scholarship.

The basic idea of analysing recurrence relations to study MSD fixed points and convergence is not new, and even in the first Bravyi and Kitaev paper which introduced MSD, the authors are using a related technique, and in https://arxiv.org/pdf/1205.3104 the authors analyse the recursion relations (in this case for qutrits), and identify fixed points and flows.

However, this paper improves upon the state of the art in the following ways:

Rather than using twirling the simplify the number of parameters in the problem, recursion relations are derived with more free parameters allowing, for example, biased-noise input states to be studied. In the first part of the paper, they allow single qubit input states to be arbitrary, so their analysis here is fully general.

The use Jacobian matrix techniques to analyse converge rates to fixed points.

In the latter part of the paper, they simplify their analysis by assuming that the state lies on the Bloch sphere equator (and hence they are now half-way back to the Bravyi-Kitaev approach which uses twirling and a single parameter), and the advantage of using 2 parameters is that 2-D flow plots can be drawn for visual analysis of the protocols.

The techniques are used to analyse a variety of MSD protocols from the literature, and in some cases unstable fixed points are identified that were not previously in the literature (although these generalise the similar fixed points found by Reichard for the Steane code, so are not so surprising)

Some intruiging fractal cases are found but not developed in any detail, and these had already been noted in [30].

Overall, this is a sound contribution, however, its novelty and timeliness are not particularly high. Magic state distillation is now a quarter of a century old, and many papers have been written on it. Recurrence relations are a standard technique to study MSDs and a paper like this could have been written 15 years ago. The editors will need to make a judgement on whether this paper has *sufficiently significant contribution to be suitable for Quantum. My own view is that the paper is currently below that threshold.*

If the authors could use their technique to derive some more impactful results, this recommendation could change.

Reply: We thank the reviewer for summarizing the main contribution of our work. We also agree with the reviewer that it's not a secret at all that MSD can be studied recursively. There are indeed many papers on MSD already that might share some similar thoughts with our work. However, the idea of recursive distillation itself is not meant to be the novelty we would like to present here. We would like to emphasize why we think our work is new to be considered as a publication:

- **We derived the analytical form for the recursive behavior of MSD within the framework of stabilizer reduction in the generic case.** MSD has been proposed for a long time, but most of the previous study on the analytical recursive behaviour is quite protocol-specific and their analysis is not easily generalizable. For example, the very first Bravyi and Kitaev paper [1] analyzed the recursive error relation for the $[[5,1,3]]$ and $[[15,1,3]]$ protocols, and Reichardt analyzed the $[[7,1,3]]$ code as well [2]. They used code-specific properties and assumed depolarizing noise to simplify their analysis. In some work they also released the depolarizing noise assumption [3–5], but their analysis is only focused on specific protocols. In our case, we generalize the analysis to almost all known MSD protocols. As long as we know the stabilizer description of the protocols, we will be able to derive the recursive relation. We also don't rely on the specific noise assumption. Therefore, our method is totally generic, and would also be applicable for any potential undiscovered MSD protocols.

- **We analyzed the existing exotic MSD protocols that are out of the current knowledge using our proposed method.** Within our framework, we analyzed these small MSD protocols proposed in [6] and studied their distillation efficiency as well as target states. Notably, we unveiled why these small MSD protocols can distill into these strange states: The angle of these exotic magic states must be a solution of a high-order polynomial equation characterized by the code structure.

- **We demonstrated the possibility for distilling new magic states by concatenating known MSD protocols. We also showed the the possibility to improve distillation efficiency by concatenating more efficient protocols.** Notably, the fractal distribution for fixed points from concatenated protocols is brand new and has not been reported in any previous literature. The fractal behavior mentioned in Rall 2017 [5] is based solely on the $[[5,1,3]]$ protocol, and they showed that fractal convergence region toward all eight Clifford-equivalent magic states. We apologize for the previous confusing statement and now revised the corresponding part as: "Notably, fractal behavior for the $[[5,1,3]]$ protocol has been studied and close input states might result in total different output states [5]. Even though the fractal behavior reported in [5] is quite different from our result, it implies the possibility of more fractal behaviors in MSD to be discovered in general."

Regarding to the comment "recurrence relations are a standard technique to study MSDs and a paper like this could have been written 15 years ago", we would like to argue that we won't be surprised if this idea has been thought by many people several years ago. However, we are surprised by the fact that no one has formalized the idea in the generic way. Moreover, As the development of recent experimental techniques, magic state distillation has attracted a lot of practical interest as well and has been demonstrated in trapped ions and neutral atoms recently [7, 8]. Therefore, we also believe our work has enough timeliness, as it facilitate the understanding of different MSD protocols in an intuitive way.

Prompted by the comment, we revised the last paragraph in the Introduction section to summarize and review the previous work that shares similar idea to emphasize the novelty of our work.

Reviewer Point 1.3 — *For example, could these techniques be used to identify an unknown MSD protocol with practical value to the quantum computing community?*

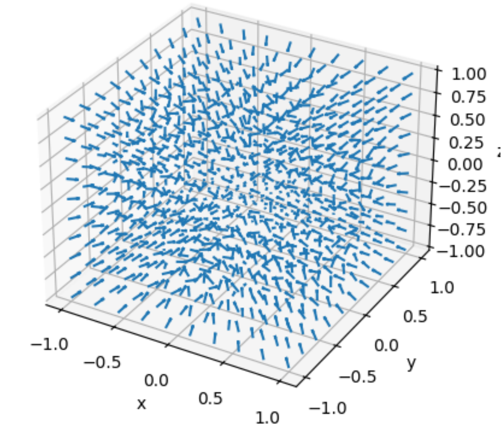
Reply: We appreciate the question raised by the reviewer and agree that this is a very good point. We actually thought about this idea when we were working on this project. Particularly, we thought about that given a particular magic state $|\theta\rangle$, how can we find new protocols that distill into it. That would be 1. identify a class of dynamical systems that share the same fixed point corresponds to $|\theta\rangle$; 2. map the dynamical systems back to the stabilizer codes (MSD protocols). The first step is not hard, but the second step is highly non-trivial as not all dynamical systems correspond to stabilizer codes. Our numerical evidence gives the feeling that there should be some hidden structure for the polynomials of the dynamical systems mapped from stabilizer codes, which we still don't understand quite well yet. Therefore, we have to firstly identify the property of the mapped polynomials that could relate to *weight enumerators* and *MacWilliams Identities* for specific stabilizer codes [9, 10]. In our opinion, this could be an interesting direction to work, but has gone beyond the scope of the current work. We think it could be a good direction for future work.

Still, our work still gives an approximate way to construct new protocols by concatenating known protocols in certain way. For example, as we showed in the Fig. 4 of the manuscript, we can generate MSD protocols for new target states by concatenating known MSD protocols that distill into different magic states. Particularly, the concatenation between the three-qubit code and the four-qubit code seems to cover the whole range for $\theta \in [0.5, 0.9]$, which demonstrate great potential for distilling target magic states directly rather than distilling $|T\rangle$ and then doing gate synthesis.

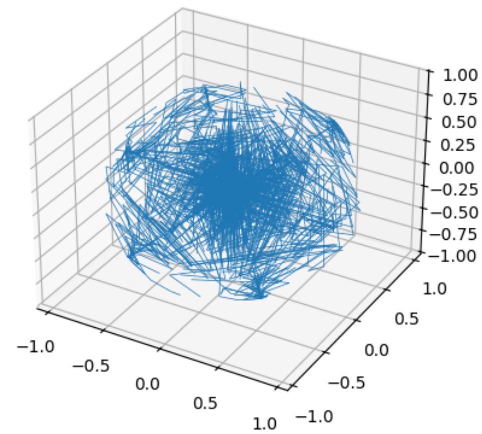
Reviewer Point 1.4 — *Another way in which the results could be strengthened is if the more results for the 3-dimensional case were given. For example, I wouldbe interested to see a full fixed point analysis and flow analysis for the full 3-d problem for the 4 MSD protocols studied. Some interestingand novel results might come from looking at the full 3-D problem with the tools developed.*

Reply: Indeed, there is no necessity to limit the analysis in the cross section for $z = 0$ within the Bloch sphere and we also have the full analytical expression for the unreduced dynamical systems. However, the bottleneck is at the visualization: There doesn't have a very good software visualization for 3D flow diagram. Common visualization software package only supports flow diagram with 2D. Even though we can still use, e.g. the 'quiver' method or plot the distillation trajectory, the plot is still not very good for intuition and might just be confusing. For example, we can plot the $[[5, 1, 3]]$ protocol in 3D as the following figures Fig. 1a and 1b:

Unfortunately, these figures are not really helpful to gain useful information or insight. Besides, as we discussed in the main context, it's a valid to assume the distillation process lies in the $z = 0$ plane: 1. Technically, when we are preparing a logical magic state $|\theta\rangle$ (non-fault-tolerantly), we will mostly suffer from logical decoherent noise as QEC code decoheres noise. We can show decoherent noise cannot create non-zero z component out of $|\theta\rangle$. 2. From a practical perspective, all known magic gate injection protocols rely on the input state to have $z = 0$. For magic states with $z \neq 0$, additional process and



(a) [[5, 1, 3]] MSD protocol plotted in 3D using the ‘pyplot.quiver‘ method.



(b) [[5, 1, 3]] MSD protocol distillation trajectory, plotted based on 1000 random sampled initial points.

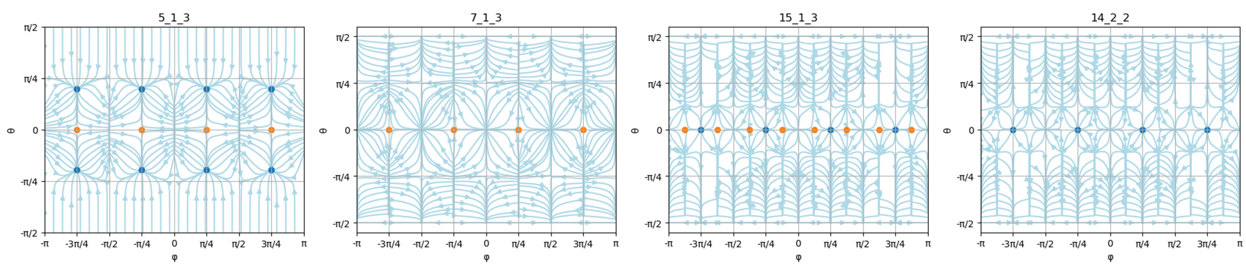


Figure 2: Flow diagram on the Bloch sphere for the four protocols using Mercator projection. Blue(yellow) dots stand for stable(unstable) fixed points for magic states.

post-selection is required to eliminate the z component [1]. Therefore, we believe it's reasonable to focus on the $z = 0$ plane, as this is the most critical and practically interesting regime we could focus on.

Surely, we also agree that it's a pity that we only present results for the $z = 0$ plane while our framework gives us the ability to do full 3D analysis. As suggested by the reviewer, we supplement our current results by two additional perspectives:

- We present the flow diagram on the Bloch sphere parametrized by the angular parameter θ, ϕ (Fig. 2). Even though states on the Bloch sphere are pure states and may not suit the realistic scenario for MSD (where the input states are noisy mixed states), this flow diagram should provide more understanding on the fixed points of the corresponding 3D dynamical systems.

- We also plot more cross-section flow diagram (See Fig. 3 and 4). Instead of setting $z = 0$, we tried different fixed value for z and also considered the cross section for $x = y$. We hope these additional cross-section 2D flow diagram can help the readers to grasp a whole understanding on the 3D behaviour.

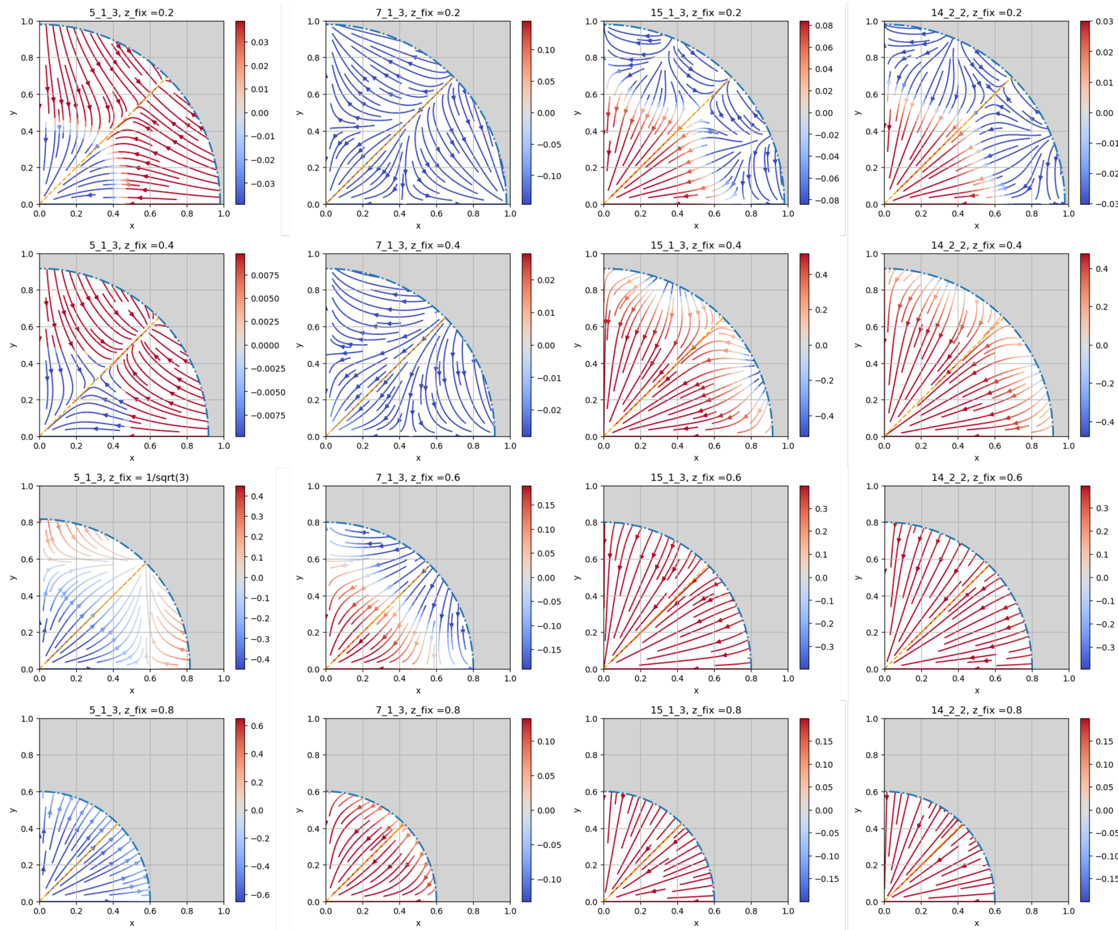


Figure 3: Flow diagram for cross section with $z \neq 0$. Colormap denotes the vertical flow direction along the z axis: Blue(red) stands for going down(up).

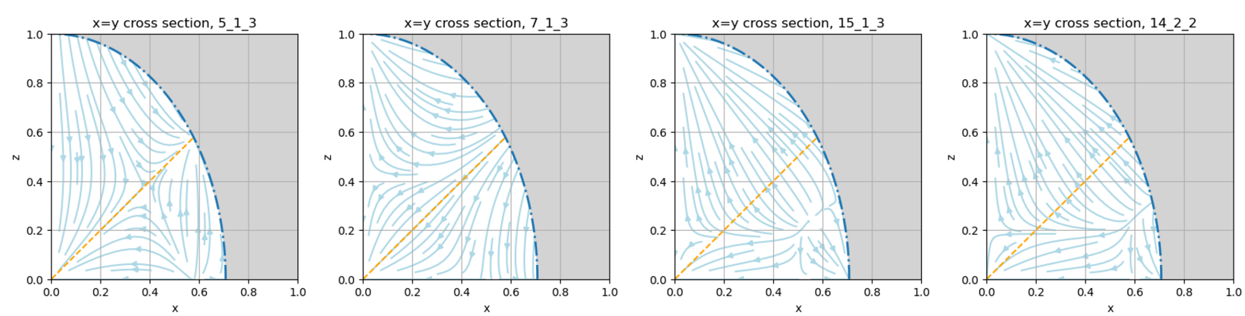


Figure 4: Flow diagram for $x = y$ cross section. The yellow dashed line denote $x = z$ for reference.

We presented these flow diagrams in the new added appendix together with result discussion and guided the readers toward the appendix in the main text.

Reviewer Point 1.5 — To what extent have you checked the technical correctness of the submission? This includes for example examining mathematical proofs or source code.

Not checked in detail, but everything looks sound.

Reply: We thank the reviewer for the positive comment on our manuscript.

Reviewer Point 1.6 — Comment on the presentation of the paper. Is it well written? Are the main results clearly laid out? Does the manuscript clearly describe assumptions and limitations? Is the literature review adequate?

The presentation is good, but there are large numbers of spelling and grammar errors.

Reply: We thank the reviewer for the positive comment on the presentation. We have revised our manuscript and correct all the typos we found.

Reviewer Point 1.7 — If the submission includes software, numerical or physical experiments, does it provide sufficient details such that they could bereproduced by readers? This includes for example source code, documentation, experimental data, experimental setupspecifications, etc.

The code used to prepare the figures is in a Github repository.

Reply: We thank the reviewer for checking our github repo.

Reviewer Point 1.8 — Suggested changes, corrections, and general comments.

There were quite a lot of typos and grammar errors. The authors should carefully check their wriiting and fix these. I think the your first open question is at least partly answered by https://arxiv.org/abs/1302.3240 (https://arxiv.org/abs/1302.3240)

Reply: We have revised our manuscript accordingly and corrected all typos we have found with our best effort.

We also thank the reviewer for sharing the reference. We are indeed aware of the known MSD protocols for $R_Z(\frac{\pi}{2^k})$ with integer k. Actually, since we already know how to construct k-orthogonal stabilizer codes with transversal $R_Z(\theta/2^k)$, we can just use these codes as the MSD protocols. However, a critical problem here is: The higher k (Clifford Hierarchy) we want, the larger the protocols will be. For example, the smallest tri-orthogonal code is $[[14, 2, 2]]$ for a 14-to-2 protocol of the $|T\rangle$ state. If we wanna find out a quadra-orthogonal code for the $\left|\sqrt{T}\right\rangle$ state, the smallest protocol will most likely be the $[[31, 1, 3]]$ punctured Reed-Muller code. The enlarged protocol size posts two critical problem: 1. The MSD protocol becomes less efficient, as we now need more raw states to distill an output state. 2. The minimum system size to achieve the MSD protocol becomes extremely large. If we are using the Steane code as the inner code, 14 logical qubits is still accessible for near-term quantum computers as it costs 14*7=98 physical qubits but 31 logical qubits require more than 200 physical qubits, and the connectivity requirement might be very demanding. Therefore, the known protocols for distilling these magic states will be less efficient and less practical.

What we would like to state in the open question is instead: **Restrict the system size**, how can we find other MSD protocols for state $|\theta\rangle$ that also have good distillation efficiency. We revised the wording in the final section accordingly to present our idea more precisely.

Reviewer 2

Reviewer Point 2.1 — ***Summary: what are the main questions posed by the manuscript and how does it answer them?*** *This manuscript investigates a unified method to analyze and visualize magic state distillation (MSD) protocols. It answers this by mapping a protocol to an iterative dynamical system, allowing for identifying distillable magic states through its fixed points and evaluating its error suppression performance using their stability. The power of the framework is demonstrated by providing the first analytical theory for"exotic" magic states, which can be related to roots of a polynomial determined by the underlying code. The authors also investigate code concatenation as a composition of these system maps, demonstrating it can generate new protocols that distill various magic states.*

Reply: We thank the reviewer for their time on reviewing and summarizing our work.

Reviewer Point 2.2 — ***What is your assessment of the paper? If you recommend acceptance, make a case that this work does indeed make a significant contribution to scholarship.*** *I recommend acceptance after minor revisions, as this work makes a significant contribution to scholarship. Its primary contribution is the systematical framework that maps MSD protocols to dynamical systems. The framework handles states with their full Bloch vector, which makes it completely general and avoids the common assumption of a specific noise model, such as depolarizing noise. Additionally, the work provides the first analytical theory for the origin of "exotic" magic states, showing their defining angles (which were previously calculated numerically) are roots of a characteristic polynomial determined by the code's structure. Moreover, the framework can be readily extended to concatenated codes, enabling a systematic search for new MSD schemes that can distill a wide range of magic states. These advances provide the community with a powerful new analytical tool and a deeper mathematical understanding of MSD.*

Reply: We are happy to know the reviewer find our work a significant contribution, and thanks the reviewer for the positive evaluation.

Reviewer Point 2.3 — ***To what extent have you checked the technical correctness of the submission? This includes for example examining mathematical proofs or source code.*** *I have carefully checked the technical correctness of the main text, except for the detailed derivation of the examples of dynamical systems(e.g., Eqs. (24)–(27)). I have also reviewed the appendix and source code available on GitHub, but have not fully verified their technical accuracy.*

Reply: We thank the reviewer for the positive comments.

Reviewer Point 2.4 — ***Comment on the presentation of the paper. Is it well written? Are the main results clearly laid out? Does the manuscript clearly describe***

assumptions and limitations? Is the literature review adequate? *The paper is generally well-written and clearly structured. The authors effectively guide the reader from the motivation and background to the core methodology and its applications through multiple sections. The main results are presented clearly with propositions and intuitive figures.*

Reply: We thank the reviewer for the positive comments on our presentation.

Reviewer Point 2.5 — ***If the submission includes software, numerical or physical experiments, does it provide sufficient details such that they could bereproduced by readers? This includes for example source code, documentation, experimental data, experimental setupspecifications, etc.*** *Their source code (https://github.com/Dran-Z/Mapping-MSD-to-Dynamical-Systems) is clear and reproducible. I tested it on my machine and was able to reproduce the results reliably.*

Reply: We thank the reviewer for their time on testing the reproducibility of our codes.

Reviewer Point 2.6 — *I have several comments on the manuscript: On Section 2 In Eq. (8), the LHS has a dependency on the logical qubit index , but the RHS does not. It would be helpful to clarify how the operation on the RHS specifically yields the state for the -th logical qubit.*

Reply: We agree the previous Eq. (8) is ambiguous. There are two ways to correct it. The first is to correct the LHS:

$$\rho^{out} = \mathrm{Tr}_{anc}[D_{\mathcal{Q}}\rho_p D_{\mathcal{Q}}^{\dagger}], \tag{1}$$

where ρ^{out} is a k-qubit state for all output states. The second is to correct the RHS, where the corrected equation should be

$$\rho_i^{out} = \mathrm{Tr}_{anc,*i}[D_{\mathcal{Q}}\rho_p D_{\mathcal{Q}}^{\dagger}]. \tag{2}$$

where the tracing out is performed for 1. all ancilla qubits. 2. all other output qubits except for the ith qubit itself. We have corrected the equation with both way and added explanation for the new equations.

Reviewer Point 2.7 — *PRX Quantum 2, 020341 (https://journals.aps.org/prxquantum/abstract/10.1103/PRX classifies MSD protocols intothree types. For instance, the typical Reed-Muller protocol is type (a) and the [[5, 1, 3]] scheme is type (b), both of which seem compatible with the stabilizer reduction formalism. It would be valuable for the authors to discuss how their framework relates to this classification,and particularly whether it applies to type (c) protocols. Such a discussion would help readers better understand the scope and generality of the proposed method within the broader landscape of MSD.*

Reply: We thank the reviewer for providing this reference. Actually, the type (c) protocols, which are based on logical Clifford measurements (e.g. in [11, 12]), are equivalent to some type (a) protocols as pointed out in [13]. Therefore, the type (c) protocols should be able to transform into type (a) protocols and then be classified as Stabilizer Reduction protocols, falling into the framework of our work.

We have added a paragraph at the end of Sec 2 to discuss the generality of Stabilizer Reduction protocols:

"Almost all known MSD protocols fall under the framework of SR. In [14], MSD protocols are classified into three types: (a) protocols based on stabilizer codes with transversal non-Clifford gates

[15, 16]; (b) protocols based on stabilizer codespace projection [6, 15, 17]; (c) protocols based on logical Clifford measurements [18, 19]. It's straightforward to incorporate the first two types to SR protocols. Even though the last type cannot be trivially regarded as SR protocols, it has been shown that these protocols are equivalent to some of the type (a) protocols [13]. Therefore, it's sufficient to consider SR protocols in the current scope of MSD."

Reviewer Point 2.8 — *On Section 3 Beginning with Proposition 3, the asterisk symbol ( ) is used (e.g.,) several times. To improve notational clarity and consistency,it may be better to either omit the symbol (e.g.,).*

Reply: We have omitted the asterisk for multiplication in the revised manuscript.

Reviewer Point 2.9 — *On page 7, the text states that the input-output error relations in Fig. 2(b) match numerical results from previous work [9, 23]. This claim would be strengthened by providing a more direct, quantitative comparison, for instance by overlaying the data from the cited works onto the plots or by quoting the relevant numerical values in the text.*

Reply: We thank the reviewer for the suggestion. However, the numerical data is not available in the origin papers. Instead we can compare the result with their analytical error relation. For example, the recursive relation for the $[[7, 1, 3]]$ protocol is given by

$$x_{out} = \frac{x^3(7 + 8x^4)}{1 + 14x^4} \tag{3}$$

for input state formed as $(x, x, 0)$ as denoted in [2]. We can show that our analytical form can be reduced to this relation by setting $x = y$. We have added explanation in the previous appendix "Analytical form of dynamical systems" to show how to arrive at the relation reported in the previous literature from our analytical form under certain noise model (The $[[7, 1, 3]]$ and $[[15, 1, 3]]$ protocol).

Reviewer Point 2.10 — *On Section 5 The fractal distribution of fixed points for concatenated protocols is an interesting finding. Does this mathematical property have a deeper physical or operational meaning for the distillation process?*

Reply: Even though the fractal distribution is indeed very interesting, we still don't know about the potential physical meaning yet. In terms of operational meaning, the fractal fixed points from concatenated protocols might prevent us from obtaining protocols distilling certain magic states. For example, we can clearly see there exist a gap in the distribution of the fixed points from Fig. 4(c) near $\theta = 0.7$. Therefore, it's very unlikely that we can get a protocol that allows to distill $|\theta = 0.7\rangle$ magic state by adding more levels of concatenation from these two base protocols. However, the fractal distribution is only presented in small codes concatenated with the 15-qubit code. It seems like when we are concatenating different small codes, the distribution can be rather uniform. As concatenating with 15-qubit code can improve the distillation efficiency, there could be some trade-off on the distillation efficiency and uniformity of the resulting fixed points. We believe more interesting work could be done based on our observation on the fractal distribution of fixed points in the future.

Reviewer Point 2.11 — *The discussion of the fractal behavior could be enhanced by briefly summarizing the findings of the cited prior work [30, 31] and then discussing more explicitly how the results of this manuscript relate to or extend those previous findings.*

Reply: We realized that our statement in the previous manuscript might be a little misleading. The fractal behaviour for fixed points of concatenated protocols in our work hasn't been discovered from any previous literature and is totally new. Instead, the fractal behaviour reported in Rall 2017 is based on the sole [[5, 1, 3]] protocol, and it shows that depending on the region of the input states, the final output states could converge to totally different target states even though the input states are very close with each other. We mentioned the work of Rall 2017 because we realize that fractal behaviour could be quite ubiquitous for MSD and would like to appeal more attention from the readers. Prompted by this comment, we revised the statement on fractal as:

"Notably, fractal behavior for the $[[5, 1, 3]]$ protocol has been studied and close input states might result in total different output states [5]. Even though the fractal behavior reported in [5] is quite different from our result, it implies the possibility of more fractal behaviors in MSD to be discovered in general."

Reviewer Point 2.12 — *The notation used in Fig. 5(a), such as "RM*k + A", could be defined more explicitly in the manuscript. While its meaning becomes clear upon inspecting the provided code, defining it in the caption or main text would improve clarity for all readers.*

Reply: We have added explanation for our notation in the caption of the corresponding figures.

Reviewer Point 2.13 — *There appears to be a subtle ambiguity in the concept of 'concatenation'. It would be helpful to distinguish between (1) a single-level MSD protocol built from a concatenated code (which has a single composite dynamical map) and (2) a multi-level scheme created by sequentially applying distinct MSD protocols (which is a more common way to use 'concatenation' in the context of MSD). It seems that the latter requires the same fixed point for each level, thus it forms a subset of the former. Figure 5(a) seems to analyze the former approach, while the cost analysis in Fig. 5(b) implicitly assumes the latter (i.e., recursively applying a single protocol to itself). Clarifying this distinction would prevent potential confusion.*

Reply: We agree that the concept of "concatenation" is a little ambiguous for MSD here. In our wording, we mostly focused on the first scenario as the case for "concatenation" because only concatenating different protocols can provide us new protocols with different fixed points. Based on our impression on previous published work on MSD, in most time people refer to the second scenario as "recursion" even though it is indeed concatenating a code with itself. As the reviewer pointed out, the second class is totally a subset of the first class.

Regarding to the Fig. (5), the (5)a stands for the input-output error relation of the concatenated new protocols. The relation denotes the behavior for executing the new concatenated protocols for **one time**. Instead, (5)b denotes the asymptotic recursive behavior of the concatenated protocols and is analyzed when we assume the protocols have been executed for **infinite times**. We have revised the caption for the Fig. (5) to emphasize the difference between these two cases.

Reviewer Point 2.14 — *The cost analysis for the [[15, 1, 3]] protocol in Fig. 5(b) is based on a specific function found in the supplementary code (*

```
N_15RM =10.2 * np.log10(1 / e) * (0.36 * np.log10(10.2 / e * np.log10(1 / e))) ** 2.46
).
```

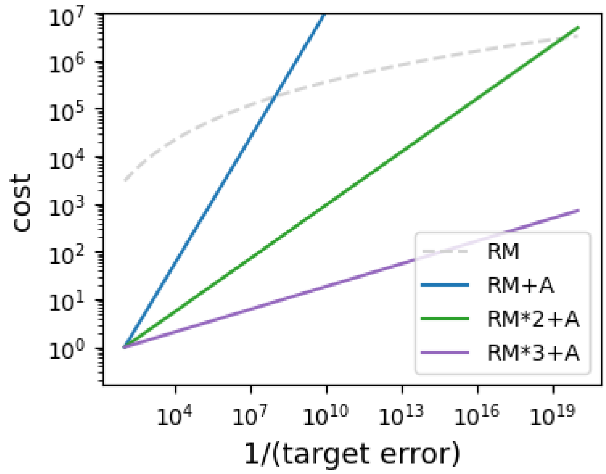


Figure 5: Revised Fig. 5(b). The dashed line is vertically lifted by a constant gap compared with the previous figure.

For completeness, itwould be beneficial to state this function explicitly in the main text and briefly explain its origin or provide a citation for it.

Reply: We thank the reviewer for noticing the specific function and we apologize for the confusion caused by this part. Here the cost of the $[[15,1,3]]$ protocol is calculated by the distillation cost multiplying the gate synthesis cost: $C = C_g \circ C_d$. The distillation cost is given by $C_d(\epsilon) \approx (\log \epsilon^{-1})^{2.46}$. The gate synthesis cost (for single-qubit gate) is given by $C_g(\epsilon) \approx 3.067 \log_2(\epsilon^{-1}) \approx 10.2 \log_{10}(\epsilon^{-1})$ as shown in [20]. Now say we'd like a gate/state with error ϵ_{tar}, we need $C_g(\epsilon_{tar})$ T gates if they are ideal. For noisy T gates, we require them to be approximately with error $\epsilon_{tar}/C_g(\epsilon_{tar})$ such that their errors, when added up in the worst case, won't be higher than our error tolerance ϵ_{tar}. Therefore, the total cost is given by

$$\begin{aligned} C &= C_g(\epsilon_{tar}) * C_d(\epsilon_{tar}/C_g(\epsilon_{tar})) \\ &= 10.2 \log_{10}(\epsilon_{tar}^{-1}) * (\log(C_g/\epsilon_{tar}))^{2.46} \\ &= 10.2 \log_{10}(\epsilon_{tar}^{-1}) * (\log(10.2 \log_{10}(\epsilon_{tar}^{-1})/\epsilon_{tar}))^{2.46} \end{aligned} \tag{4}$$

It's worth mention that we indeed made a mistake in our previous calculation and we underestimated the cost for the $[[15,1,3]]$ protocol by a constant factor, as $\log a = \log_{10} a * \log 10 \approx 2.30 * \log_{10} a$ rather than $0.36 * \log_{10} a$. However, this doesn't affect the main conclusion. The revised figure will look like as Fig. 5. We have revised our simulation code and added explanation for cost analysis in the corersponding main text.

Reviewer Point 2.15 — *The final sentence of the caption for Fig. 5 appears to be cut off ("With more concatenation with the...").*

Reply: The sentence should be "With more concatenation with the 'RM' protocols, we can reduce the overall cost for distillation." and we have corrected in the revised manuscript.

Reviewer Point 2.16 — *Other Comments The analysis throughout the manuscript assumes perfect Clifford operations. It would be valuable to add a brief discussion on how the presence of*

Clifford errors might affect the framework and the performance of MSD protocols. In particular, for protocols with only linearerror suppression, it seems plausible that even a small amount of Clifford noise could negate the distillation effect entirely. A comment on this possibility would be insightful.

Reply: This is a very important point. People often assumes perfect Clifford operations in MSD and most work on MSD (even for many recent work) are based on this assumption. However, it won't be such a case in practical quantum computer. Sadly, the discussion on the impact of noisy Clifford operations is still very scarce. The imperfection on Clifford gates in MSD has been discussed in [4] but not in a generic way. The imperfection on measurements is discussed in our recent work [21], where we found out that the presence of imperfect measurements can break up the high-order distillation efficiency and degrade it to linear, which can incur exponentially higher distillation cost. However, we still lack a good unified understanding on MSD with noisy Clifford operations. Even though we assumed the perfect Clifford operations in this work, we believe the dynamical system mapping can also be used to study the effect of noisy Clifford gates and measurements.

We have added a paragraph as another open question in the last section of main text to comment on this direction.

Reviewer Point 2.17 — *I found several typos: "distilltion", "appraoching", "supporession", and "procotols".*

Reply: We have checked the whole manuscript once again and corrected all typos we found.

Again, we thank the reviewer for these really detailed comments, which implies the reviewer's careful effort on reviewing the manuscript and we believe the reviewer must have spent quite a lot time on reading and checking our work. These comments greatly helped us improve the quality of our manuscript. If the editorial policy allows, we would like to acknowledge the anonymous reviewer's help at the end of of revised manuscript.

References

[1] S. Bravyi and A. Kitaev, "Universal quantum computation with ideal clifford gates and noisy ancillas", Physical Review A—Atomic, Molecular, and Optical Physics **71**, 022316 (2005).

[2] B. W. Reichardt, "Improved magic states distillation for quantum universality", arXiv preprint quant-ph/0411036 (2004).

[3] B. W. Reichardt, "Improved magic states distillation for quantum universality", Quantum Information Processing **4**, 251–264 (2005).

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